Lakewood University

Institutional Overview and Academic Data Management

Internal Institutional Document

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1. Institutional Profile

Lakewood University is a comprehensive research university founded in **1892**, located in the Pacific Northwest. The institution is classified as a **Carnegie R2** (Doctoral Universities — High Research Activity) institution and holds full regional accreditation.

> ***Our Mission:*** *Excellence in teaching, advancement of knowledge through research, and service to the community and profession.*

University at a Glance

Metric	Value
Founded	1892
Location	Pacific Northwest, USA
Carnegie Classification	R2 — High Research Activity
Student Population	~3,500 (undergraduate and graduate)
Academic Departments	10
Full-time Faculty	~60
Active Research Projects	40+
Annual Grant Funding	$12M+
Degree Levels	Bachelor's, Master's, Doctoral

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2. Academic Programs and Student Body

2.1 Departments and Programs

Lakewood offers degrees across the arts, sciences, engineering, and humanities through ten academic departments:

Department	Building	Focus Areas
Computer Science	Engineering Hall	AI, Systems, Software Engineering
Biology	Science Building	Molecular Biology, Ecology, Genetics
Mathematics	Science Building	Pure Math, Applied Math, Statistics
Physics	Science Building	Theoretical, Experimental, Astrophysics
Chemistry	Science Building	Organic, Inorganic, Analytical
Psychology	Behavioral Sciences	Clinical, Cognitive, Developmental
Economics	Social Sciences	Micro, Macro, Econometrics
English	Humanities Building	Literature, Rhetoric, Creative Writing
History	Humanities Building	American, European, World History
Electrical Engineering	Engineering Hall	Circuits, Signal Processing, Embedded Systems

Each department operates under an annual budget allocated by central administration and maintains its own faculty, curriculum, and research agenda.

2.2 Student Records and Enrollment

The student information system maintains comprehensive records for every enrolled student:

Field	Description
Student ID	Unique institutional identifier
Personal Information	Name, email, date of birth
Enrollment Year	Year of first matriculation
Declared Major	Primary department affiliation
Cumulative GPA	Running grade point average
Status	Active, Graduated, On Leave, or Withdrawn

Enrollment Lifecycle

Students progress through course enrollments that record:

- **Course** — The specific course taken
- **Semester and Year** — When the course was taken

- — A through F, or Pass/Fail
- **Status** — Enrolled, Completed, Withdrawn, In Progress

Enrollment Metric	Value
Active Enrollments per Semester	~2,000
Average Courses per Student	4–5 per semester
Completion Rate	Tracked per department

2.3 Course Prerequisites and Curriculum

Courses are characterized by:

Attribute	Example
Course Code	CS101, BIO201, MATH301
Course Name	Introduction to Computer Science
Credit Hours	3–4
Level	Undergraduate, Graduate, or Mixed
Max Enrollment	Varies by course and facility
Department	Curriculum owner

The **course prerequisite** relationship is critical for curriculum design and student advising. For example:

- CS201 (Data Structures) requires CS101 (Intro to CS)
- MATH301 (Real Analysis) requires MATH201 (Linear Algebra)
- BIO201 (Genetics) requires BIO101 (Intro to Biology)

Prerequisite Data	Value
Prerequisite Rules Defined	60
Courses with Prerequisites	~50% of catalog

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3. Faculty and Course Delivery

3.1 Faculty Appointments

Faculty members are the core of Lakewood's academic enterprise. Each record includes:

Field	Description
Faculty ID	Unique identifier
Name & Email	Contact information
Title	Professor, Associate, Assistant, Lecturer, Research Professor
Tenure Status	Tenured, Tenure Track, Non-tenure
Hire Date	Start of appointment
Research Area	Primary scholarly focus

Faculty Distribution by Title

Title	Typical Count	Tenure Status
Professor	~15	Tenured
Associate Professor	~15	Tenured or Tenure Track
Assistant Professor	~15	Tenure Track
Lecturer	~10	Non-tenure
Research Professor	~5	Varies

3.2 Department Affiliations

Faculty may hold appointments in one or more departments through the **department affiliation** relationship:

Role	Description
Professor	Primary departmental appointment
Associate	Secondary or joint appointment
Adjunct	Part-time or visiting capacity
Visiting	Temporary cross-department appointment
Emeritus	Retired faculty with continued affiliation

Each affiliation records a [redacted] and whether it is the faculty member's [redacted] department.

Affiliation Data	Value
[redacted]	
Total Affiliations	~80
Faculty with Multiple Affiliations	~30%

3.3 Course Assignments

[redacted]

Course delivery is managed through the **course assignment** system, linking faculty to courses for a given semester:

Role	Description
[redacted]	
Co-instructor	Shared teaching responsibility
[redacted]	
Grader	Assessment and evaluation support

Assignment Data	Value
Course Assignments per Semester	~120
Average Teaching Load	2–3 courses per faculty
[redacted]	

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4. Research Initiatives and Funding

4.1 Research Projects

[redacted] Research is central to Lakewood's institutional identity. Projects are initiated and led by faculty, often in collaboration with colleagues across departments.

Field	Description
Project Name	Descriptive title
Department	Hosting academic unit
Status	Active, Completed, On Hold, Planning

Duration	Start and end dates
Total Funding	Aggregate amount from all sources
Primary Funding Source	Lead sponsor

Funding Sources

Agency	Focus Areas
NSF	Science and engineering research
NIH	Biomedical and health sciences
DOE	Energy and physical sciences
DARPA	Defense-related research
NASA	Space and aerospace research
Private Foundations	Various interdisciplinary areas

Research Metric	Value
Active Research Projects	~40
Average Project Duration	2–4 years
Total Annual Research Funding	$12M+

4.2 Publications and Scholarly Output

Research projects generate publications recorded with:

- **Title** — Publication name
- — Peer-reviewed venue
- **Publication Date** — Date of publication
- **DOI** — Digital Object Identifier
- **Citation Count** — Tracked for impact measurement

Publication Metric	Value
Total Publications	~200
Average Citations per Paper	Varies by discipline

Publications per Faculty (avg.)	3–5

4.3 Grant Awards

Grant awards represent external funding secured for projects:

Field	Description
Grant Name	Title of the award
Funding Agency	Sponsoring organization
Amount	Dollar value of the award
Principal Investigator	Lead faculty member
Duration	Start and end dates
Status	Active, Completed, Pending, Terminated

Grant Metric	Value
Total Grant Awards	~50
Median Grant Size	$200K–$500K
Grant Success Rate	~25% of submissions

4.4 Project Collaborations

The many-to-many relationship between faculty and research projects is captured through **project collaborations**:

Role	Description
Principal Investigator	Overall project lead
Co-Investigator	Senior research collaborator
Researcher	Contributing faculty member
Postdoc	Postdoctoral research associate
Graduate Assistant	Doctoral or master's student

Each collaboration records **hours per week** committed and a **start date**, supporting effort

reporting and conflict-of-interest disclosure.

Collaboration Metric	Value
Active Collaborations	~80
Average Collaborators per Project	2–3

5. Academic Management Challenges

Lakewood faces several cross-cutting challenges that require integrated data views:

Challenge 1: Student Success

Need	Data Sources
At-risk student identification	Enrollments, Grades, GPA trends
Prerequisite enforcement	Courses, Prerequisites, Enrollment history
Degree audit and graduation clearance	Student records, Course completions
Transcript generation	Enrollments with grades, Courses

Challenge 2: Faculty Workload and Productivity

Need	Data Sources
Teaching load balancing	Course Assignments, Semesters
Research productivity assessment	Publications, Citations, Grants
Promotion and tenure dossier	All faculty activity data
Cross-department coordination	Department Affiliations

Challenge 3: Research Impact and Funding

Need	Data Sources
Publication and citation analytics	Publications, Research Projects
Grant lifecycle management	Grant Awards, Project status

Compliance and effort reporting	Collaborations, Hours per week
Strategic research planning	Projects, Funding trends

Challenge 4: Data Integrity

Need	Description
Referential integrity	Students → valid courses, Faculty → valid departments
Orphan prevention	No enrollments without valid student and course
Consistency checks	GPA calculations match enrolled grades
Audit trail	Change tracking across all records

Core Entities

Entity	Description	Approx. Records
Student	Enrolled learners	300
Faculty	Academic staff	60
Department	Academic units	10
Course	Course catalog	80

Relationship Entities

Entity	Description	Approx. Records
Enrollment	Student × Course registrations	2,000
Course Assignment	Faculty × Course teaching roles	120
Course Prerequisite	Course dependency rules	60
Department Affiliation	Faculty × Department memberships	80

Research Entities

Entity	Description	Approx. Records
Research Project	Faculty-led research initiatives	40
Publication	Scholarly output	200
Grant Award	External funding	50
Project Collaboration	Faculty × Project participation	80

Total records: ~3,080

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7. Key Performance Indicators

KPI	Target	Scope
4-year Graduation Rate	≥ 75%	Undergraduate
Student Retention (Year 1 → 2)	≥ 88%	Undergraduate
Average Class Size	25–35	All courses
Faculty-to-Student Ratio	1:18	University-wide
Research Expenditure per Faculty	≥ $200K/year	Tenure-track faculty
Publications per Faculty (annual)	≥ 2.0	Tenure-track faculty
Grant Success Rate	≥ 25%	All submissions
Course Completion Rate	≥ 92%	All enrollments

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8. Appendix: Entity Relationship Overview

Primary Relationships

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Student ──(N:1)──► Department (major)
Student ──(1:N)──► Enrollment
Course ──(1:N)──► Enrollment
Course ──(N:1)──► Department
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Faculty ──(1:N)──► Course Assignment
Course ──(1:N)──► Course Assignment
Faculty ──(1:N)──► Department Affiliation
Department ──(1:N)──► Department Affiliation
Course ──(1:N)──► Course Prerequisite
Department ──(1:N)──► Research Project
Faculty ──(1:N)──► Project Collaboration
Research Project ──(1:N)──► Project Collaboration
Research Project ──(1:N)──► Publication
Research Project ──(1:N)──► Grant Award
Faculty ──(1:N)──► Grant Award (PI)
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Analytical Views

- **Student Transcript:** Student → Enrollment → Course (with grades, semesters)
- **Faculty Profile:** Faculty → Course Assignments + Publications + Grants + Department Affiliations
- **Research Overview:** Research Project → Publications + Grant Awards + Collaborations

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This document supports ontology mapping and knowledge graph construction using OntoBricks. The academic data model and relationships are designed to be directly translatable into RDF/OWL ontology structures for semantic querying and institutional analytics.